

SUBHASISH MUKHERJEE

C# / .NET CONSULTANT | ASP.NET CORE WEB API | CLEAN ARCHITECTURE

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PROFESSIONAL SUMMARY

Senior software engineer with 20+ years of enterprise application experience across C#, ASP.NET/ASP.NET Core, Java/Spring Boot, PHP/Laravel, Node.js and SQL. Hands-on .NET 7 project experience includes production-grade REST APIs for UAE fintech and digital-banking platforms using Clean Architecture, Entity Framework Core, SQL Server, JWT, Azure AD B2C, Docker, Azure DevOps and AKS. Strong in API design, SOLID, dependency injection, security, database performance, external-system integration, CI/CD and production troubleshooting. Recent AI engineering work in RAG, GraphRAG, LLM applications and agent/tooling adds a modern differentiator to a deep backend and architecture profile.

TECHNICAL SKILLS

C# / .NET: C# 11, .NET 7, ASP.NET Core Web API, OOP, SOLID, generics, LINQ, async/await, exception handling

Architecture: Clean/Onion Architecture, layered services, dependency inversion, dependency injection, DTOs/contracts, repository pattern, CQRS concepts, microservices

Data: Entity Framework Core 7, SQL Server, Azure SQL, relational modeling, transactions, LINQ, indexing, query optimization, performance tuning

Security: JWT Bearer, Azure AD B2C, Microsoft Graph SDK, OAuth 2.0, RBAC, device validation, KYC controls, secure configuration

API & Integration: REST, Swagger/OpenAPI, API versioning, ESB/middleware, UAE Pass, PixLab, Firebase FCM, HTTP integrations

Azure / DevOps: Azure, AKS, ACR, Docker, Kubernetes manifests, Azure DevOps YAML, Application Insights, multi-environment deployment

Testing & Engineering: xUnit, Postman, JMeter, Git, GitHub, CI/CD, code review, logging, audit trails, vulnerability scanning, production debugging

AI Engineering: RAG, hybrid retrieval, pgvector, PostgreSQL FTS, GraphRAG, LLM applications, prompt engineering, Ollama/local LLMs, agent/tooling

SELECTED .NET PROJECT EXPERIENCE

InvestNation - UAE FinTech Investment Platform | ASP.NET Core 7 | C# 11 | EF Core 7 | SQL Server | Azure

Production mobile backend for retail and corporate investment customers, implemented as a Clean Architecture REST API and deployed on Azure Kubernetes Service.

- Built REST API modules for authentication, customer onboarding, accounts, investments, risk profiling, referrals, payments, configuration and back-office operations.
- Worked across API, Application, Contract, Domain and Infrastructure layers using interface-driven services and dependency injection.
- Implemented stateful onboarding with email/mobile OTP, Emirates ID OCR, KYC document handling, risk assessment and configurable retry/cooling rules.
- Integrated JWT Bearer, Azure AD B2C / Microsoft Graph, UAE Pass OAuth 2.0, device registration and multi-provider identity flows.
- Implemented investment subscription/redemption, daily limits, core-banking middleware orchestration and Quartz.NET scheduled referral processing.
- Used Swagger/OpenAPI, API versioning, xUnit, Audit.NET, HTTP logging and Application Insights; supported Docker and Azure DevOps CI/CD into ACR/AKS.

Dhabi Finance House - UAE Digital Banking Platform | ASP.NET Core 7 | C# | EF Core 7 | SQL Server | Azure

Digital-banking backend for retail and corporate customers covering onboarding, KYC, authentication, accounts, documents and operational services.

- Developed REST API capabilities in a Clean Architecture solution spanning Domain, Application, Infrastructure, Contract and API layers.
- Implemented multi-step onboarding covering OTP, Azure B2C registration, customer validation, address/income/tax capture, Emirates ID/Passport upload and re-KYC flows.
- Integrated JWT Bearer authentication with Azure AD B2C, Microsoft Graph and device validation; enforced KYC expiry and role-specific login flows.
- Built account and configuration APIs for CIF data, balances, statements, multilingual content and mobile-application configuration.
- Integrated ESB communication services, PixLab document utilities, Audit.NET and Application Insights; supported Docker, ACR, AKS and Azure DevOps pipelines.

PROFESSIONAL EXPERIENCE

Senior Developer | Agropreneurs Market Technologies Pvt. Ltd. | Mar 2025 - Feb 2026

- Maintained and optimized enterprise backend services and APIs in a cloud-hosted product environment, with emphasis on reliability, performance and production issue resolution.
- Worked across backend, database and frontend boundaries using Node.js, GraphQL, React, Git and Agile delivery practices.
- Applied AI-assisted development using modern LLM tools to accelerate prototyping, debugging and automated testing.

Senior Technical Developer / Architect | Qolaris Data India Pvt. Ltd. | Jan 2023 - Feb 2025

- Architected scalable backends and REST APIs for multi-tenant property search, School Management, eLearning and HRMS platforms.

- Applied layered architecture, database optimization, automated testing and structural refactoring; documented work improved query/runtime efficiency by 25%.
- Used AI-assisted engineering and prompt orchestration for rapid prototyping, debugging, documentation and test generation.

Lead Full-Stack Developer | Eastern India Technosoft Pvt. Ltd. | Apr 2021 - Dec 2022

- Architected backend and microservice solutions using Java Spring Boot/MySQL and delivered secure enterprise REST APIs.
- Worked on code reviews, security hardening, performance tuning and deployment-pipeline troubleshooting in Agile/Scrum teams.

Senior Software Engineer | Ambrossia Infotech Solutions Pvt. Ltd. | Feb 2020 - Mar 2021

- Designed relational data structures and optimized complex SQL across MySQL and Microsoft SQL Server for high-traffic systems.
- Managed testing, code reviews, system specifications and integration work with security-focused engineering practices.

Senior Developer / Technical Lead | Matrix Media Solutions Pvt. Ltd. | May 2014 - Jan 2020

- Delivered enterprise applications across Laravel, CodeIgniter, CakePHP and WordPress; provided technical leadership and database performance tuning.
- Built secure backend integrations and services for multiple client accounts.

Software Engineer | Elegant Systems Powerweb Pvt. Ltd. | Jan 2004 - Apr 2014

- Engineered enterprise web systems using PHP, ASP, JSP and MySQL, including lead management, booking management and prepaid sales-chain applications.
- Earlier hands-on ASP.NET/C# and SQL Server work included online tutoring, patient/lead management, online magazine publishing and medical transcription systems.

EARLIER ASP.NET / C# PROJECT HIGHLIGHTS

- MYETutor CA Inc. - ASP.NET/C#/SQL Server online tutoring platform with tutor/student management and timezone-aware scheduling; Lead & Patient Management System with lead distribution, RBAC, surgery scheduling and patient tracking.
- Medical Transcription Application - ASP.NET/C#/SQL Server system with audio management, policy-driven assignment and patient/transcription tracking.

RECENT AI ENGINEERING & OPEN-SOURCE PORTFOLIO

- Enterprise RAG Assistant - Laravel + PostgreSQL/pgvector + Gemini/Ollama; hybrid retrieval, context optimization and citation grounding. GraphRAG / Advanced RAG - Python/React graph-based retrieval exploration.
- TinyGPT - Python/PyTorch transformer and LLM fundamentals project; LocalOps AI - autonomous developer-agent project covering tool discovery, execution, reasoning, memory and testing. GitHub: <https://github.com/msubhasish1601>

EDUCATION & CREDENTIALS

- Master of Computer Applications (MCA) - Sikkim Manipal University, 2008
- Bachelor of Science (B.Sc.), Chemistry - Barkatullah University, 2001
- Scrum Fundamental Certified (SFC) - SCRUMstudy, 2021
- AI Tools Certification - Claude, Gemini, ChatGPT, Perplexity, DeepSeek and related AI productivity tools, 2026